Vikas Nagar

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PROFESSIONAL SUMMARY

Associate Data Scientist with advanced training, hands-on project and internship experience in Python-based data analysis, deep exploratory data analysis (EDA), statistical analysis, and machine learning. Proficient in Pandas, NumPy, SQL, and Scikit-learn for cleaning, transforming, and modeling large structured datasets (700K+ records), including trend and seasonality analysis. Built interactive Streamlit dashboards and reproducible end-to-end pipelines; communicates insights clearly to technical and non-technical stakeholders.

TECHNICAL SKILLS

Programming & Databases: Python, SQL, Microsoft SQL Server

Data Analytics & Statistics: Exploratory Data Analysis (EDA), Data Cleaning & Preprocessing, Data Transformation, Feature Engineering, Statistical Analysis, Hypothesis Testing, Correlation Analysis, Regression Analysis, Trend & Seasonality Analysis

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Time Series Forecasting, Market Basket Analysis, Predictive Modeling, Model Evaluation, Hyperparameter Tuning, MLOps Fundamentals

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, SHAP, Matplotlib, Seaborn, SciPy, Statsmodels, MLxtend

Tools & Deployment: Git, GitHub, Streamlit, Jupyter Notebook, VS Code, Excel, Power BI, Microsoft SQL Server

EXPERIENCE

Data Science Intern | Pantech ProEd Pvt Ltd *2023, 2024 & 2026*

Python, Data Analytics, Machine Learning

- Completed three structured internship programs covering Python programming, data analytics, and machine learning through hands-on technical exercises and practical projects.
- Applied Python, data preprocessing, EDA, visualization, and feature engineering to structured datasets; built and evaluated ML models using standard training and performance-evaluation workflows.

PROJECTS

E-Commerce Customer Analytics Platform | GitHub *May 2026 - Jul 2026*

Python, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, SHAP, Streamlit, Git

- Performed EDA on 742K+ transactions, 22K customers, and 42K products; validated data quality and engineered 85+ features (RFM, category affinity, loyalty metrics) into a 118-column Customer360 dataset.
- Designed RFM-based customer segmentation, classifying users into High Spenders, Loyal, At-Risk, and Churned segments; framed these as clear modeling targets for downstream classification.
- Built a modular, leakage-proof Scikit-learn pipeline (preprocessing, feature generation, training, evaluation, inference); benchmarked 6 classification models with cross-validation and hyperparameter tuning, reaching 98.9% AUC for High-Spender prediction and 95.6% AUC for Churn detection.
- Explained predictions with SHAP and built an interactive Streamlit dashboard; found the top 2.75% of customers generate 18.1x more revenue than the bottom 76.8%.

Telecom Customer Behavioral Segmentation | GitHub *Jan 2025 - Mar 2025*

Python, Pandas, NumPy, Matplotlib, Scikit-learn, K-Means, MLxtend, Streamlit, Git

- Engineered 7 behavioral features from raw telecom data; performed EDA, correlation and VIF analysis to validate data quality and feature suitability for downstream clustering.
- Segmented 1,000+ users into 6 behavioral groups by benchmarking K-Means, Agglomerative Clustering, and DBSCAN; selected K-Means based on Elbow Method and Silhouette Score validation.
- Identified a high-value segment (3% of users, 5.1x higher income, 7.7x higher loyalty) and applied Apriori association rule mining across 13 services, surfacing cross-sell bundles with 91% confidence and 3.07 lift.
- Documented a reproducible pipeline, automated batch scoring, and built a Streamlit dashboard for KPI visualization and team reporting.

Insurance Claims EDA and Hypothesis Testing | GitHub *Oct 2026 - Nov 2026*

Python, Pandas, NumPy, SciPy, Statsmodels, Matplotlib, Seaborn

- Merged claims and customer demographics, then cleaned data types, corrected date-of-birth errors, deduplicated to the latest claim per customer, and imputed missing values.
- Ran five hypothesis tests at 95% confidence (difference, independence, association, Spearman correlation), finding 2018 claims significantly above a $10,000 baseline; analyzed monthly claim trends and fraud patterns by segment.

Movie Recommendation System (CineMatch) | GitHub *Jun 2025 - Jul 2025*

Python, Pandas, Scikit-learn, Streamlit, OMDb API, Git

- Built a content-based recommendation engine on 7,692 MovieLens titles using 19 genre and year features with cosine similarity to generate Top-5 recommendations.
- Cleaned 1,113 malformed titles, filtered records to 1985-2018 for data quality, and deployed an interactive Streamlit application with real-time OMDb API integration, delivering posters, ratings, and instant recommendations.

CERTIFICATIONS & ACHIEVEMENTS

- **Advanced Certification in Data Science & AI** - E&ICT Academy, IIT Guwahati *Aug 2024 - May 2025*
- **Certified Data Scientist** - AnalytixLabs *May 2025*
- **Certified Data Analyst (Python, Excel, SQL & Power BI)** - AnalytixLabs *Nov 2024*
- **Data Science with AI Training** - Internshala Trainings, NSDC *Sep 2023*

EDUCATION

B.Tech in Computer Science Engineering *2021 - 2025*

Modi Institute of Technology, Kota | CGPA: 8.2/10.0

Final Year Project: Credit Risk Analysis - Probability of Default Model with 88% accuracy, minimizing banking credit risk